From force diagrams to motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a force diagram to predict acceleration and velocity change.

SUCCESS CRITERIA - I can...

1. Write a force sum along each chosen axis.
2. Connect net force and mass to acceleration.
3. Use velocity direction to decide the initial speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “parallel” and “perpendicular” to explain your model or result. Sentence starters are optional.

parallel: along the same direction as a line or surface

perpendicular: at a right angle, 90°, to a line or surface

component: the part of a vector along one chosen axis

01 Warm-up: bring back an earlier idea

A 3 kg cart has 5 N right and 26 N left.
Find acceleration. What else must you know to decide if it speeds up?

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

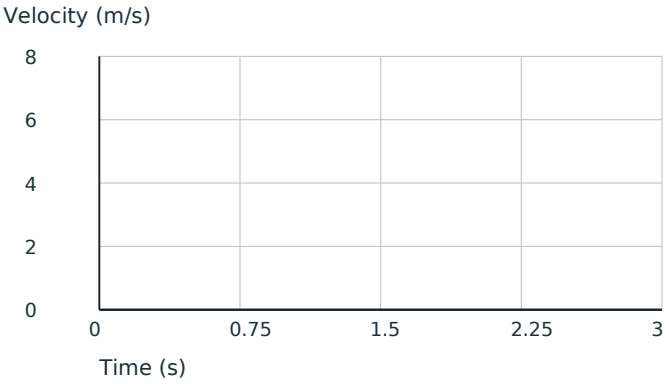
02 Worked example: explain the model

Reference: $F_{g,\ parallel} = mg \sin\theta$; $F_{g,\ perpendicular} = mg \cos\theta$
$\Sigma F = ma$.
Ramp angle θ above horizontal. Components replace the weight vector; they are not extra forces.

Worked reasoning: A 2 kg cart initially at rest has a constant 4 N net force right for 3 s. Acceleration is 2 m/s²; final speed is 6 m/s right.

Your turn: Draw a force diagram and a velocity-time graph that agree.
Mark where mass enters the reasoning.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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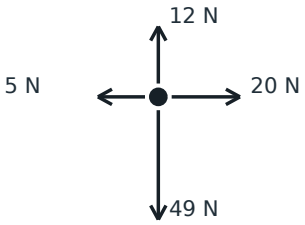
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Angled pull

A 5 kg box stays on a level floor. A rope pulls with components 20 N right and 12 N up; friction is 5 N left.
Find normal force and horizontal acceleration.
Use g=9.8.

Sentence starter: I found ... [unit]. This value means ... in this situation.



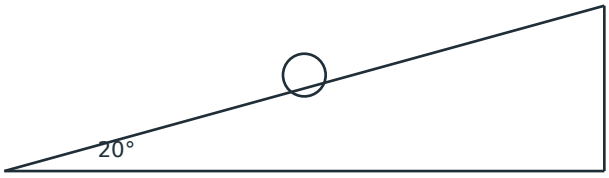
The dot represents the object. Normal not shown; find it

Marzano check - circle: 1 2 3

04 Frictionless ramp

A 2 kg cart slides down a 20° frictionless ramp.
Find normal force and acceleration along the ramp.
Use g=9.8 m/s².

Sentence starter: I found ... [unit]. This value means ... in this situation.



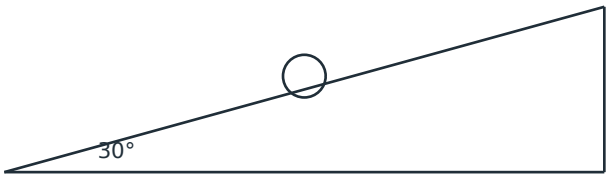
The marker represents the object on the sloping surface.

Marzano check - circle: 1 2 3

05 Oppose upward motion

A 4 kg block slides up a 30° ramp after release. Kinetic friction is 5 N down the ramp.
Use g = 9.8 m/s² and choose up the ramp as positive.
Complete the force diagram, calculate acceleration, and describe the initial change in speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

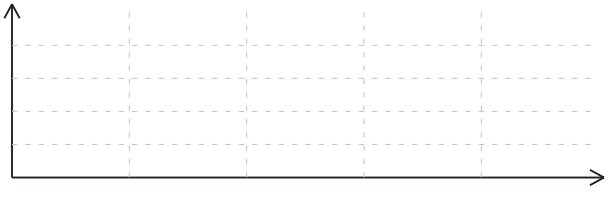
06 Review: connect an earlier idea

Velocity is +5 m/s for 7 s, then -4 m/s for 5 s.
Find displacement and distance.
Explain the reference information needed for an angle of 35°.
Sketch the velocity-time model; label the given values.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Velocity (m/s)



Time (s)

Marzano check - circle: 1 2 3

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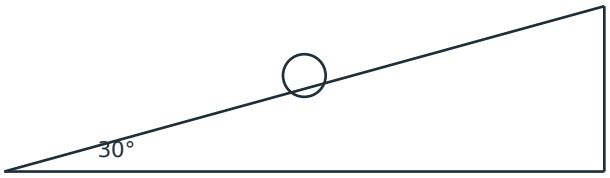
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 2 kg block slides down a 30° ramp with 3.8 N friction up the ramp.
Find along-ramp acceleration and normal force.
Use g=9.8.

Sentence starter: I found ... [unit]. This value means ... in this situation.



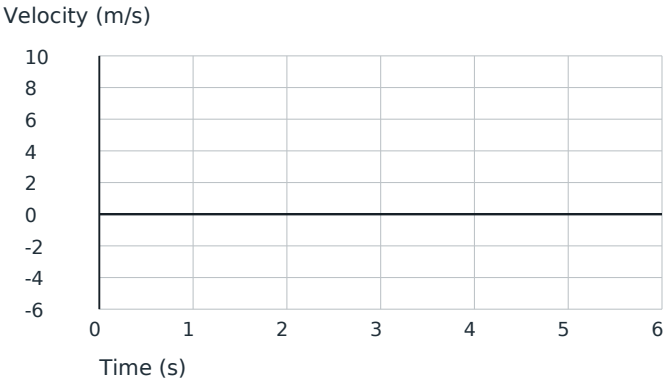
2 kg block slides down; friction 3.8 N up the ramp.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart moving right at 8 m/s has a constant 4 N net force left for 6 s.
Find when it turns, final velocity, displacement and distance.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3